⚡ QUICK REFERENCE - ECOHUB QR SCANNER

🚀 KHỞI ĐỘNG NHANH

Development

```
python app.py
→ http://127.0.0.1:5000
```

Production

```
EcoHub_QR_Scanner.exe
→ Auto open browser
```

🎯 WORKFLOW

```
1. Quét QR      → AI detect
2. Auto record  → Đóng gói
3. Stop record  → Save video
4. Auto upload  → S3 storage
5. Cooldown 5m  → Ready
```

🔧 CÀI ĐẶT LẦN ĐẦU

Camera

```
⚙ Cài đặt camera
→ Test Camera
→ Start Camera (nếu OK)
→ Cấu hình USB/RTSP
→ Áp dụng
```

S3

```
☁ Cài đặt S3
→ Nhập credentials
→ Test kết nối
→ Lưu cấu hình
```

📍 MENU NAVIGATION

```
🏠 Dashboard
   - Camera live feed
   - QR code hiện tại
   - Quay video
   - Thông tin đơn

⚙ Cài đặt camera
   - Test/Start/Stop camera
   - Cấu hình camera (max 2)
   - Độ nhạy AI
   - Auto-record toggle

📦 Kho lưu trữ
   - Tab: Video trên S3
   - Tab: Lịch sử upload

☁ Cài đặt S3
   - S3 credentials
   - Test connection
```

⌨ SHORTCUTS

Không có keyboard shortcuts. Sử dụng mouse/touch.

🔑 CONFIG FILES

Development

```
eco_hub_demo/
├── config.json     Configuration
├── config.key      Encryption key
└── videos/         Local videos
```

Production

```
C:\Users\<user>\EcoHub_QR_Scanner\
├── config.json
├── config.key
└── videos\
```

📊 STATUS INDICATORS

Camera Badge

```
✅ Đang chạy    = Camera ON
⚪ Chưa khởi động = Camera OFF
```

Recording Status

```
🔴 Recording     = Đang quay
⚪ Idle          = Không quay
```

Upload Status

```
⏳ Pending       = Chưa upload
⬆️ Uploading     = Đang upload (%)
✅ Success       = Upload thành công
❌ Failed        = Upload lỗi
```

🐛 XỬ LÝ LỖI NHANH

Camera không khởi động

```
1. Cắm lại USB
2. Đóng Skype/Zoom
3. Restart app
```

S3 upload failed

```
1. Check credentials (Cài đặt S3)
2. Test connection
3. Check IAM permissions
```

Video bị lag

```
# Đã fix - RTSP buffer = 0
# Nếu vẫn lag: Giảm FPS
```

Templates not found

```
# Rebuild
pyinstaller build.spec --clean
```

📞 CONSOLE LOGS

Normal

```
[STARTUP] Config loaded
[CAMERA DEBUG] fps=30, queue=1/2
[AI DEBUG] fps=5.0, t_total=0.189s
[UPLOAD WORKER] Success: video.mp4
```

Errors

```
[ERROR] Cannot open camera
[ERROR] S3 upload failed
[ERROR] Templates not found
```

🔢 DEFAULT VALUES

```
Camera FPS:     30 (USB), 25 (RTSP)
AI FPS:         5
Recording FPS:  15
Resolution:     1280x720
Cooldown:       300s (5 minutes)
Buffer:         90 frames
Upload chunk:   10 MB
Upload threshold: 50 MB
```

📦 BUILD COMMANDS

Build EXE

```
build_exe.bat
# hoặc
pyinstaller build.spec --clean
```

Build Installer

```
build_installer.bat
# hoặc
"C:\Program Files (x86)\Inno Setup 6\ISCC.exe" setup.iss
```

Clean Build

```
rmdir /s /q dist build
```

🔐 SECURITY

Encrypted Fields

```
- S3 Access Key
- S3 Secret Key
```

Encryption Method

```
Algorithm: Fernet (symmetric)
Key size:  44 bytes (base64)
Location:  config.key
```

🎛️ TUNING PARAMETERS

Camera

```
# camera_manager.py
width = 1280
height = 720
fps = 30
buffer_size = 0  # RTSP only
```

AI Scanner

```
# ai_scanner.py
TARGET_FPS = 5
COOLDOWN_SECONDS = 300
RESIZE_WIDTH = 480
```

Recorder

```
# recorder.py
RECORDING_FPS = 15.0
BUFFER_SIZE = 90
```

S3 Upload

```
# s3_service.py
MULTIPART_THRESHOLD = 50 * 1024 * 1024  # 50 MB
MULTIPART_CHUNKSIZE = 10 * 1024 * 1024  # 10 MB
```

📡 API ENDPOINTS

```
GET  /                  Login
GET  /dashboard         Dashboard
GET  /camera-settings   Camera settings
GET  /storage-settings  S3 settings
GET  /storage           Storage page
GET  /video_feed        MJPEG stream
GET  /status            Realtime status
GET  /upload-status     Upload status

POST /test_camera       Test camera
POST /start_cameras     Start cameras
POST /stop_cameras      Stop cameras
POST /start_recording   Start recording
POST /stop_recording    Stop recording
POST /reset_order       Reset order code
```

🔄 THREADING MODEL

```
1. Main Thread          Flask HTTP (port 5000)
2. Camera Thread 1-2    Read frames (10ms)
3. AI Thread 1-2        Scan QR (200ms)
4. Recorder Thread      Write video
5. Upload Thread        S3 upload worker
```

📏 LIMITS

```
Max cameras:        2
Max resolution:     1920x1080
Min resolution:     640x480
Max FPS:            30
Upload file size:   No limit (multipart)
```

```
Queue size:         Unlimited (disk space)
Cooldown:           5 minutes
Upload history:     1 day (auto-cleanup)
```

🌐 BROWSER SUPPORT

```
✅ Chrome/Edge      Recommended
✅ Firefox          OK
⚠️ Safari           Limited MJPEG
```

📞 SUPPORT

Documentation: - README.md - Quick start - DOCS.md - Full documentation - BUILD_GUIDE.md - Build instructions

Logs:

```
# Dev
python app.py

# Prod
EcoHub_QR_Scanner.exe
```

Config:

```
Development: eco_hub_demo/config.json
Production:  C:\Users\<user>\EcoHub_QR_Scanner\config.json
```

Version: 1.0.0
Updated: 09/02/2026